

Kathmandu Engineering College
Department of Electronics, Communication and Information



An Introduction About Signals

Lab Report 1

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1 Objective

- Observe and analyze the generation of continuous and discrete-time output signals via simulation software.

2 Theory: Signal

- A signal is a physical phenomenon and can be represented in many ways. For example: physical, electrical, optical, electromagnetic, etc.
- A signal is defined as any physical quantity that varies with space, time, or any other independent variables.
- Mathematically, we can describe the signal as a function of one or more independent variables.
- Real-world continuous examples:
 - $X_1(t) = 2t$ (varies linearly with time)
 - $X_2(t) = 5t^2 + 6t$ (varies quadratically)
 - $I(x, y) = 8x^2 + 56xy + 4y^2$ (two independent variables x and y)
- Independent variables can be continuous or discrete.
- A signal can either be real-valued or complex-valued.

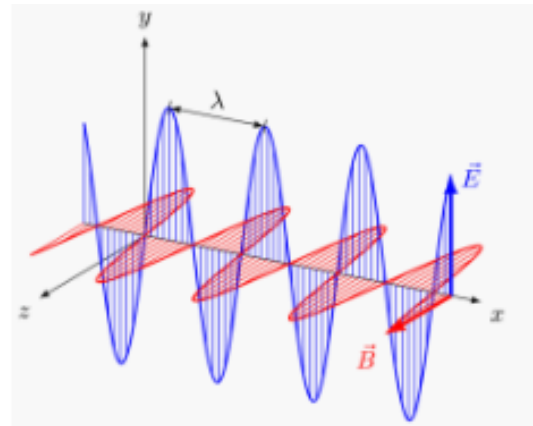


Figure 1: Hardware block diagram layout.

2.1 Classification of Signal Based on the Basis of Function

2.1.1 Continuous-Time Signal

- A continuous-time signal is a signal where the values of the function are continuous over the entire period of time t .
- That is, a signal $X(t)$ is said to be a continuous-time signal if it is mathematically defined for all time constraints across the continuous interval t .

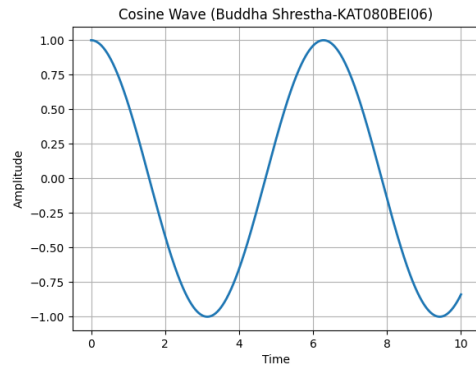


Figure 2: Continuous-time Cosine signal plot.

2.1.2 Discrete-Time Signal

- A discrete-time signal is a signal which is explicitly defined only at discrete instants of time.
- The independent variable possesses discrete value sequences only, which are usually uniformly spaced sample points.
- A discrete-time signal is represented as $X[n]$ or sometimes by $x(n)$.

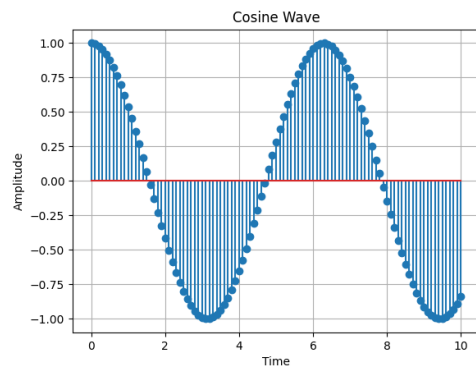


Figure 3: Discrete-time Cosine signal plot.

3 Source Code & Simulation Output Layouts

3.1 Sine Signal

The sine waveform can be generated by defining a finely sampled time vector over a chosen interval. By applying the sine function to this sequence, we obtain a smooth and continuous sinusoidal curve that represents periodic oscillations over time.

```
n = 0:0.01:10;  
y1 = sin(n);  
subplot(4,2,1)  
plot(n, y1, 'linewidth', 2)  
title('Sine Wave (Buddha Shrestha -  
      KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

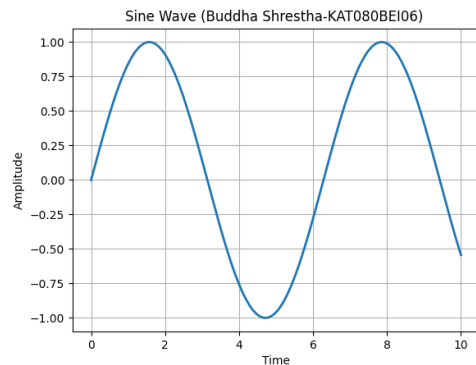


Figure 4: Sine Wave Plot.

3.2 Cosine Signal

A cosine signal is a periodic waveform that oscillates between positive and negative values, similar to a sine wave but shifted in phase. Mathematically, it is expressed as the cosine function of time, producing a smooth curve that starts at its maximum amplitude when time is zero.

```
y2 = cos(n);  
subplot(4,2,2)  
plot(n, y2, 'linewidth', 2)  
title('Cosine Wave (Buddha Shrestha -  
      KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

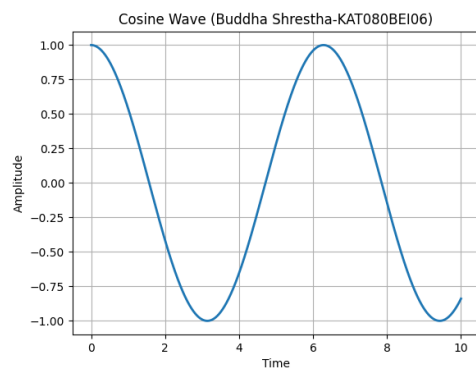


Figure 5: Cosine Wave Plot.

3.3 Square Signal

A square wave is a non-sinusoidal periodic waveform that alternates abruptly between two amplitude levels, typically $+1$ and -1 . Unlike sine or cosine waves, which vary smoothly, the square wave switches instantaneously, making it useful in digital electronics and switching circuits.

```
Fs = 1000;
t = 0:(1/Fs):1;
f = 10;
y3 = sign(sin(2*pi*f*t));
subplot(4,2,3)
plot(t, y3)
ylim([-1.5 1.5])
title('Square wave (Buddha Shrestha-KAT080BEI06)')
ylabel('Amplitude')
xlabel('Time')
```

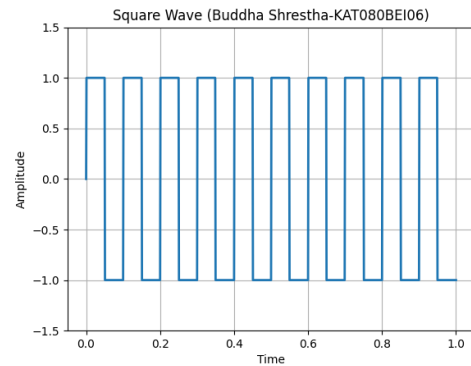


Figure 6: Square Wave Plot.

3.4 Sawtooth Signal

A saw tooth signal is a type of periodic waveform that rises linearly over time and then sharply drops back to its starting value, repeating this pattern continuously. It is commonly used in signal processing and electronics to represent non-sinusoidal periodic behavior.

```
y4 = sawtooth(2*pi*f*t);
subplot(4,2,4)
plot(t, y4)
title('SawTooth (Buddha Shrestha-KAT080BEI06)')
ylabel('Amplitude')
xlabel('Time')
```

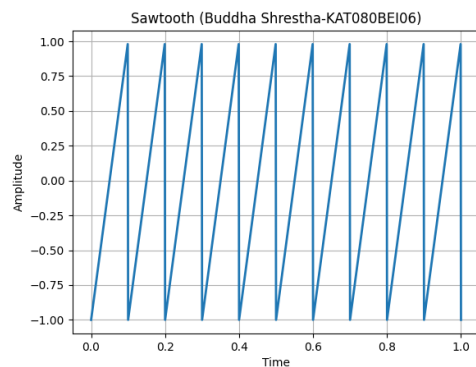


Figure 7: Sawtooth Wave Plot.

3.5 Exponential Signal

An exponential signal is a waveform whose amplitude grows or decays exponentially with time. It is widely used in modeling natural processes such as population growth, radioactive decay, and charging/discharging in electrical circuits. Mathematically, it is expressed as an exponential function of time.

```
x = 0:0.01:10;  
expo = exp(t);  
subplot(4,1,3)  
plot(t, expo)  
title('Exponential (Buddha Shrestha-KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

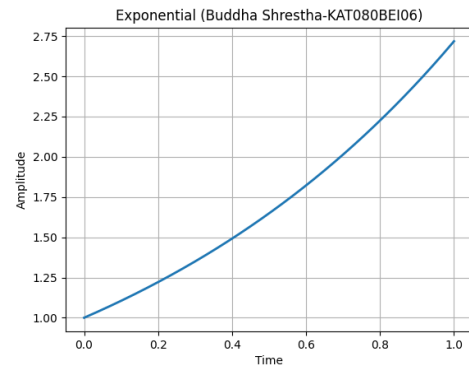


Figure 8: Exponential Wave Plot.

3.6 Unit Step

A unit step signal is a basic discrete-time or continuous-time function that remains zero for all negative values of time and becomes one for all values of time greater than or equal to zero. It is widely used in system analysis to represent sudden changes or switching behavior.

```
t = -5:0.01:5;  
y = t>=0;  
subplot(4,2,5)  
plot(t, y, 'linewidth', 2)  
ylim([-1.5 1.5])  
title('Unit Step (Buddha Shrestha-KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

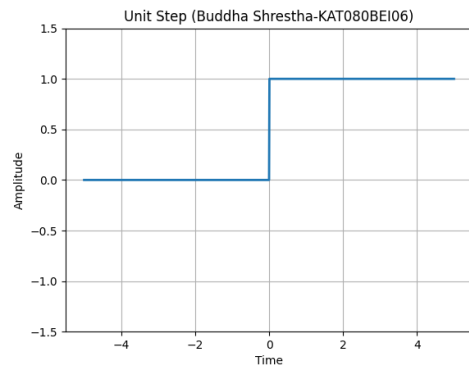


Figure 9: Unit Step Plot.

3.7 Unit Ramp

A unit ramp signal is a basic function that increases linearly with time. It remains zero for negative values of time and starts rising with a slope of one from time zero onward. This signal is often used in system analysis to represent a steadily increasing input.

```
y = t.*(t>0);  
subplot(4,2,6)  
plot(t, y, 'linewidth', 2)  
ylim([-1.5 1.5])  
title('Unit Ramp (Buddha Shrestha-KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

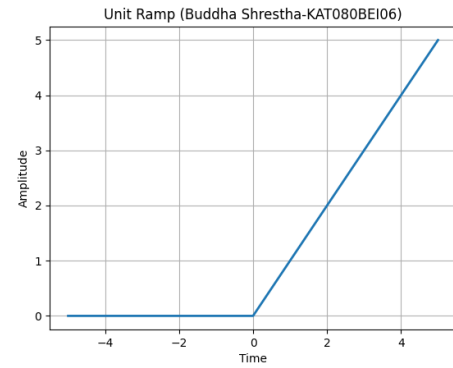


Figure 10: Unit Ramp Plot.

3.8 Rectangular

A rectangular pulse signal is a waveform that maintains a constant amplitude for a fixed duration and then returns to zero. It is often used in digital communication systems to represent a finite burst of energy or a block of data transmission.

```
y = abs(t)<=1;  
subplot(4,2,6)  
plot(t, y, 'linewidth', 2)  
ylim([-1.5 1.5])  
title('Rectangular (Buddha Shrestha-KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')
```

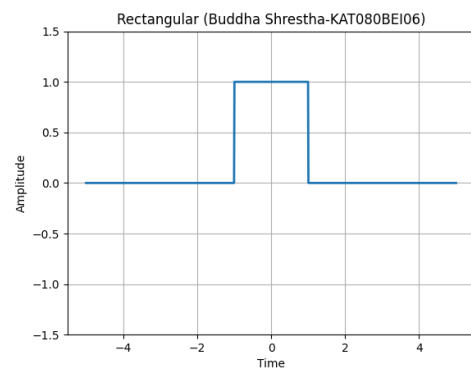


Figure 11: Rectangular Plot.

3.9 Sinc

A sinc signal is a mathematical function defined as $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$. It is widely used in signal processing, especially in Fourier analysis and sampling theory, because it represents the ideal low-pass filter and appears naturally in reconstruction of band-limited signals. The *sinc* function starts at 1 when $t=0$ and oscillates with decreasing amplitude as t moves away from zero.

```
y = sinc(t);  
subplot(4,1,4)  
plot(t, y, 'linewidth', 2)  
title('Sinc (Buddha Shrestha-KAT080BEI06)')  
ylabel('Amplitude')  
xlabel('Time')  
grid on
```

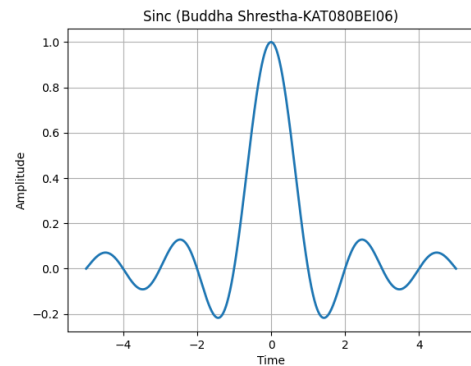


Figure 12: Sinc Wave Plot.

4 Discussion

This experiment demonstrates how MATLAB can be used to visualize key continuous and discrete signals essential to signal processing and system analysis:

- **Sine/Cosine Waves:** Continuous, periodic signals fundamental to Fourier analysis and natural oscillations.
- **Sawtooth/Square Waves:** Non-sinusoidal periodic signals with sharp transitions, ideal for electronics and waveform synthesis.
- **Exponential Signals:** Modeled on natural growth or decay processes, such as capacitor charging.
- **Unit Step/Ramp Signals:** Standard inputs for system analysis, representing sudden switching or steady increases.
- **Rectangular Pulse:** A finite-duration signal primarily used in digital data transmission.
- **Sinc Function:** A vital signal in sampling theory and ideal low-pass filtering.

Ultimately, mastering these foundational waveforms in MATLAB facilitates the study of advanced concepts like modulation, filtering, and system response.

5 Conclusion

This study successfully utilized MATLAB to mathematically define and visualize key fundamental signals: sine, cosine, sawtooth, square, exponential, unit step, ramp, rectangular, and sinc.

The practical application underscored their distinct roles in system analysis and communication theory:

- **Periodic Signals** (Sine, cosine, sawtooth, square) illustrated characteristic oscillatory behavior.
- **Non-Periodic Signals** (Exponential, step, ramp) effectively modeled real-world processes like growth, switching, and linear increases.
- **Communication Signals** (Rectangular, sinc) highlighted critical concepts in digital transmission and sampling theory.

Ultimately, the experiment provided a foundational understanding of signal behavior essential for advanced engineering applications.